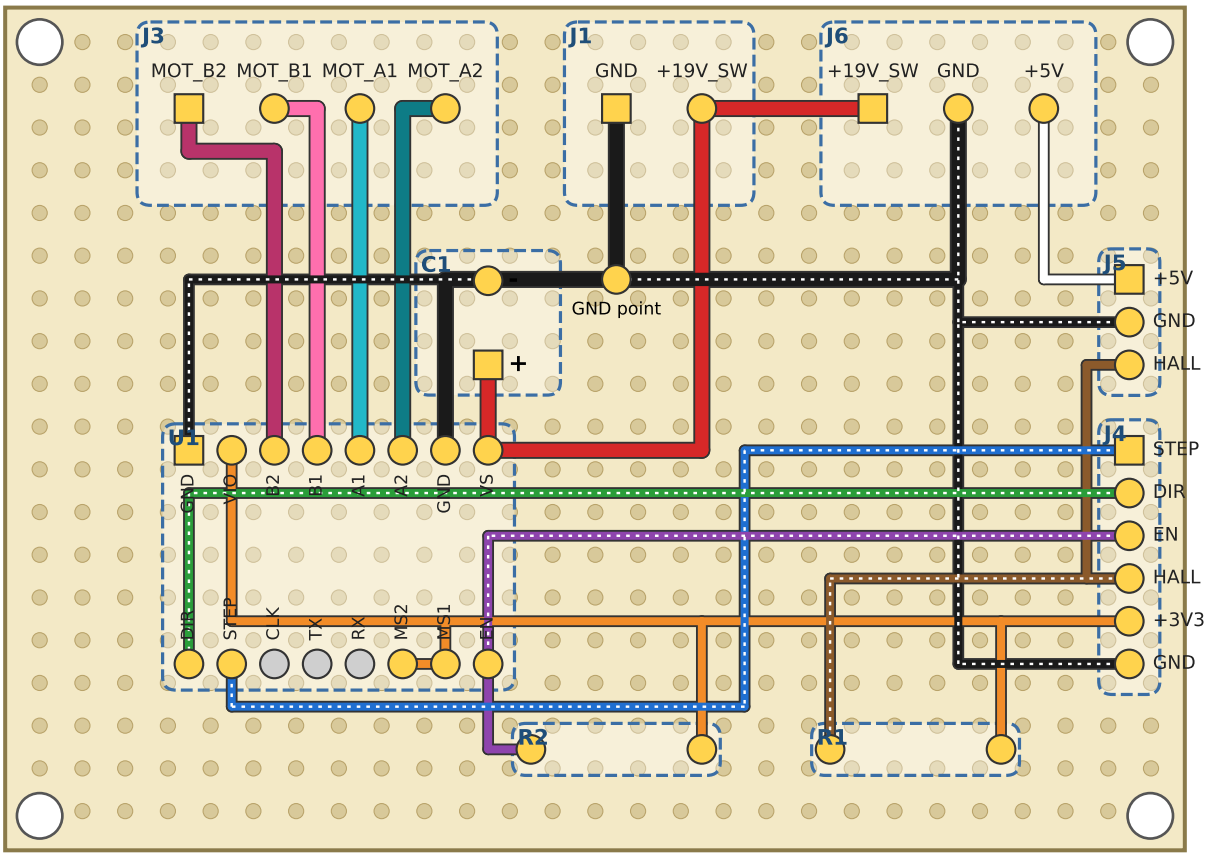
Piper-Watch Level 2 perfboard (70 x 50) - BOTTOM VIEW (solder side)



BOTTOM VIEW - the solder side: the board turned over left-to-right,
so the back of the case is still at the top. Wire it as you see it here.

Parts are on the other side (dashed outlines); J4 / J5 are now on the RIGHT.

Thick = heavy wire (19 V, GND, motor).

Dotted stripe = INSULATED wire (it crosses another run); the rest can be bare.

Grey pads are not connected. TMC2209 square pad = pin 1 (GND).

The same board from the parts side: perfboard_wiring.pdf

C1 100uF 35V (at U1 VS)

J1 19V in: + - right to left from this side

J3 motor: A2 A1 B1 B2 (right to left from this side)

J4 to ESP32: STEP DIR EN HALL 3V3 GND

J5 to Hall: 5V GND OUT

J6 to buck: 19V, GND, 5V

R1 10k (Hall pull-up)

R2 10k (opt., EN pull-up)

U1 BTT TMC2209

- 19 V (switched)
- GND
- 5 V (Hall)
- 3V3 (from the ESP32)
- STEP
- DIR
- EN
- HALL
- coil A2
- coil A1
- coil B1
- coil B2